ITO4132 Introduction to Databases

Assignment 2 - SQL

Generative AI tools **cannot be used in this assessment task**

In this assessment, you must not use generative artificial intelligence (AI) to generate any materials or content in relation to the assessment task.

City Run (CR) is a running carnival which is held on separate dates at various capital cities in Australia during different seasons (Summer, Autumn, Winter and Spring) of the year. The carnival naming convention that City Run uses is *CR <season name> Series <city name> <year>*. So, for example, a carnival to be held during the Summer season in Melbourne in 2024 will be named as *CR Summer Series Melbourne 2024.*

Anyone can attend a CR Carnival, the carnivals are open to all members of the public. A carnival is run on a particular date, in a particular location and only lasts for one day. CR ***only runs one carnival on any particular date***. During a carnival a range of events are offered from the following list (only some may be offered):

- Marathon 42.2 Km
- Half Marathon 21.1 Km
- 10 Km Run
- 5 Km Run
- 3 Km Community Run/Walk

City Run expects to offer around 10 - 20 such events across all carnivals in a given year.

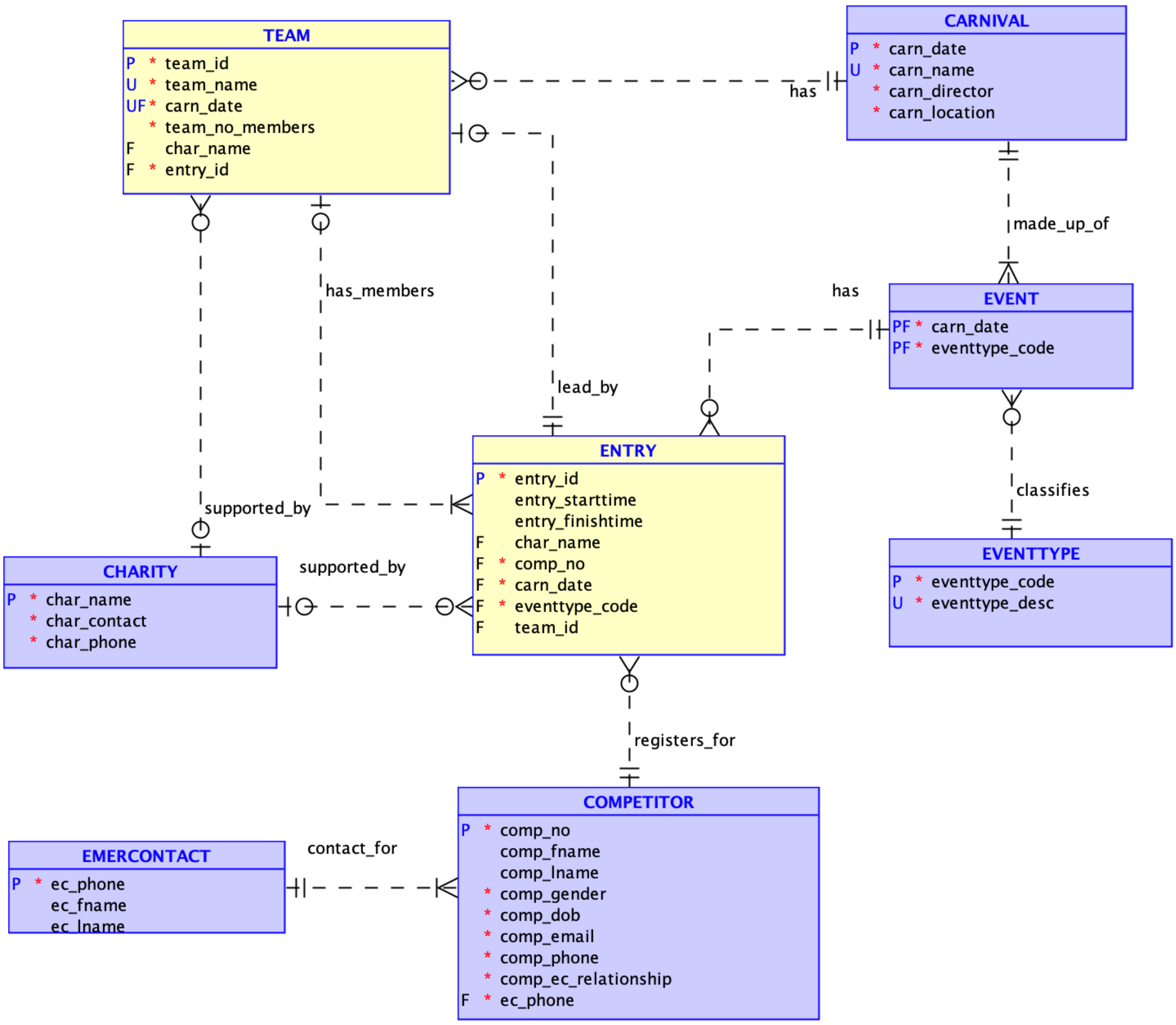
When a competitor initially registers for City Run, they are assigned a unique competitor number. A competitor is required to provide details of an emergency contact at the time of registration. The relationship to the competitor can be Parent (P), Guardian (G), Partner (T) or Friend (F).

When a carnival is being offered, City Run contacts all registered competitors and provides details of the carnival date and what events are on offer. Competitors can enter for only one event within a carnival. Every entry is assigned a unique entry id (e.g. 3021). Using official timing devices at the carnival, City Run records the entrants starting and finishing times.

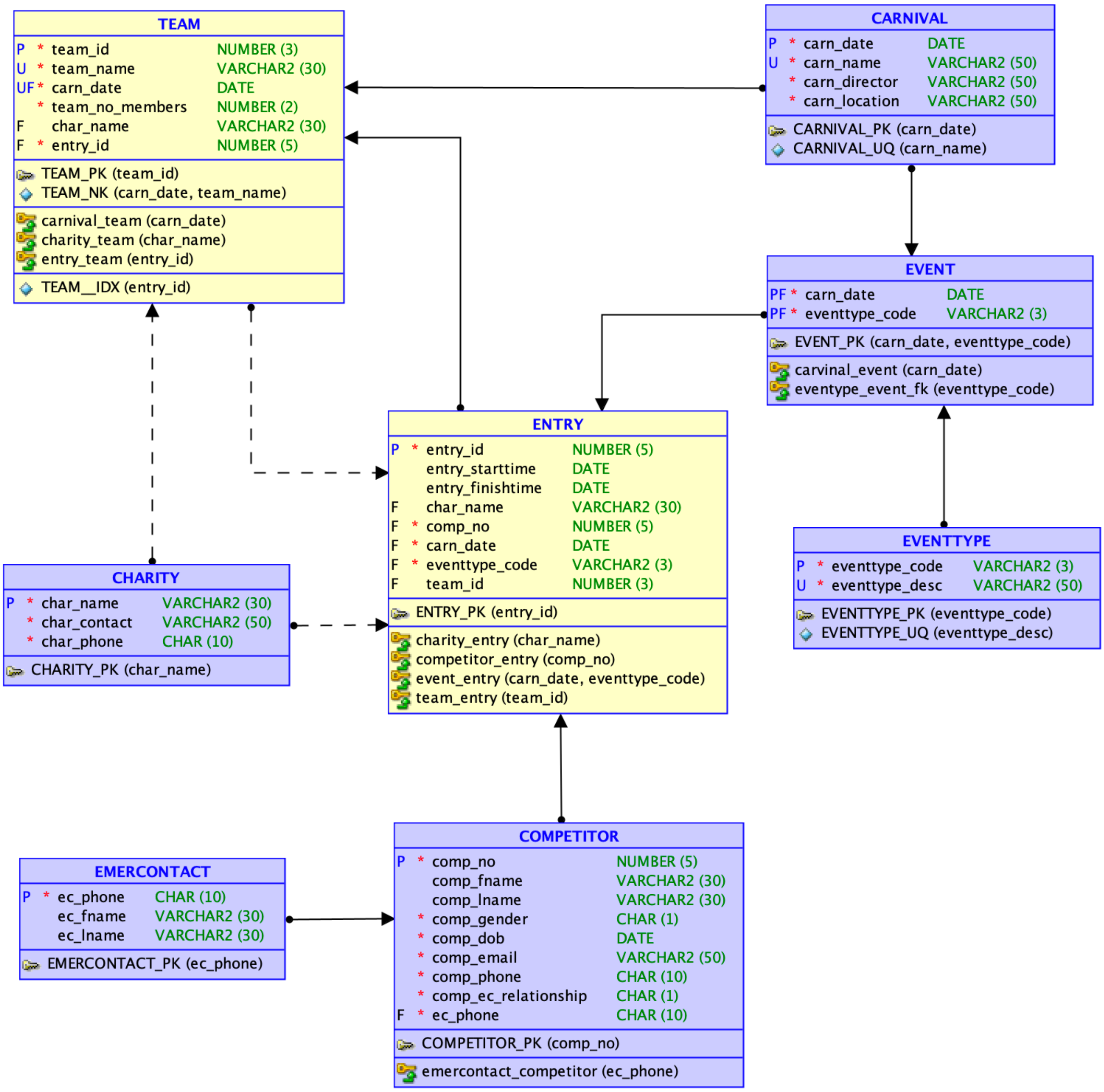
A major focus of the City Run Carnivals is to raise funds for various charities. When a competitor enters an event, they may nominate a charity for which they will raise funds (not all competitors will select a charity for each event they enter). Competitors who have entered the carnival can also form teams with other competitors, whom they know and who have entered the carnival, to support their training and run as a group. The first competitor to register a team for a given carnival is assigned as the Team Manager. Teams are identified by a unique team name which the team manager must select when they first create the team. This team manager can then add/invite other competitors from the carnival to join their team. Team names are unique *only* within a given carnival, A given team name may be reused by different competitors in a different carnival as teams are recreated for each carnival depending on which competitors have entered an event for the carnival. City Run wishes to record, as part of the stored data, how many members are on each team. Teams may also nominate a charity for which they will raise funds, although not all teams will do so. All charities for which funds can be raised must first be approved by City Run.

Note that an individual competitor may be supporting a charity as an individual and also the same or a different charity as a team member.

A data model has been developed for this situation. The logical model is shown below:



From this logical model the following relational model has been created:



For this assignment, you will populate these tables with appropriate test data and complete the tasks specified below. **You must ensure that any activities you need to carry out in the database to complete the assignment tasks conform to the requirements of the provided data model**.

The schema/insert file for creating this model (cr-schema-insert.sql) is available in the archive ass2-student.zip - this file creates the City Run tables and populates several of the tables (those shown in purple on the supplied model) - you should read this schema carefully and be sure you understand the various data requirements. **You must not alter the schema file in any manner, it must be used as supplied**. This schema file contains a single commit after the inserts have completed since this is setting up an initial state of the database for you to work with, you should not use this as your standard method of approach.

IMPORTANT points for you to observe, when completing this assignment, are:

1. The ass2-student.zip archive also contains several files for you to type your answers in, **you should ensure these files are regularly pushed to GitLab server so a clear development history is available for the marker to verify (a minimum of fourteen pushes are required - two per task).** In each file, you ***must*** fill in the header details with your name and student ID before beginning any work. Your SQL script files **must not include any SPOOL or ECHO commands**. Although you might include such commands when testing your work **they must be removed before submission** (a grade penalty will be applied if your documents contain spool or echo commands).

2. You are free to make assumptions if needed. However, ***your assumptions must align with the details here and in the Ed Assignment 2 forum*** and must be clearly documented (see the required submission files).
REMEMBER you must keep up to date with the Ed Assignment 2 forum where further clarifications may be posted (this forum is to be treated as your client). **Please be careful to ensure you do not post anything which includes your reasoning, logic or any part of your work to this assignment forum as doing so violates Monash plagiarism/collusion rules.**

3. SQL views ***must not*** be used in arriving at any solutions for the tasks you are required to complete as part of this assessment.

4. In handling dates, the default date format must not be assumed; ***you must make use of the TO_DATE and TO_CHAR functions where appropriate***.

5. **ANSI joins must be used** where joining of tables in SQL is required.

6. In completing the following tasks, you must ***design your test data so that you always get output for the SQL scripts/queries specified below*** - this may require you to add further data as you move through completing the required tasks. Such extra data MUST be added as part of Task 2 (i.e. as part of your load of test data). ***Queries that are correct but do not produce any output ("no rows selected" message) using your test data will lose 50% of the marks allocated***, so you should carefully check your test data and ensure it thoroughly validates your SQL queries.

****

In answering these tasks you are ONLY permitted to use the SQL structures and syntax which have been covered within this unit:

- **Module 2A The Relational Model**
- **Module 3B Creating and Altering the Database Structure (DDL),**
- **Module 4 DML & Transactions and SQL Part I,**
- **Module 5 SQL Part II and III and the Oracle Common Functions document.**
- **Module 6 Non-Relational Databases**

SQL syntax and commands outside of the covered work, as detailed above, will NOT be accepted/marked.

****

Assignment Tasks

TASK 1: Relational Database Queries - Relational Algebra (6 marks)

Your answers for this task (Task 1) must be written in an Ms Word document or Google document. Once you have completed all questions, download or print the document as a pdf file and name the file as **T1-cr-ra.pdf**.

For this task you are required to write the **relational algebra operations** for the following queries (your answer must show an understanding of query efficiency). *Please ensure you copy and paste each of the questions below (a) - (c) into your answer document*.

List of symbols for copying/pasting as you enter your answers below:

project: π, select: σ, join: ⋈, intersect: ∩, union: ∪, minus: -

Example:
Question:

List the Team ID and Team name for all teams involved with the carnival held on the 1st Feb 2023 .

Answer:

R = π team_id, team_name (σ carn_date= "01-Feb-2023" TEAM)

OR

R1 = σ carn_date = "01-Feb-2023" TEAM

R = π team_id, team_name (R1)

(a) List the carnival date, carnival name name and location for all carnivals which have an event type code of "10K".

[1 mark]

(b) List competitor number, first and last name and date of birth for any competitor who is registered by City Run but has as yet not registered for an events (added an entry)

[2 marks]

(c) List competitor number, first and last name, and their emergency contact’s first name, last name and phone number for all competitors who registered in a carnival named "CR Summer Series Sydney 2023"'.

[3 marks]

TASK 2: Data Insert (10 marks)

You may need to rerun the schema (cr-schema-insert.sql), especially when you have been experimenting with your solutions and may have corrupted the database unintentionally. If you suspect that there might be such problems, simply rerun the schema. The schema includes the appropriate drop commands at the head of the file.

Load selected tables with your own additional test data: using the supplied **T2-cr-insert.sql** script file, code the SQL commands which will insert, as a minimum, the following sample data into the yellow coloured relations in the logical model -

- 20 ENTRIES,
- 5 TEAMS

Please note, these are the *minimum number of entries you must insert*; you are encouraged to insert more to provide a richer data set to draw from and to validate your queries in task 4.

For this task **only**, data that you add in the database should follow the rules mentioned below:

1. *You may treat all of the data that you add as a single transaction since you are setting up the initial test state for the database.*
2. The primary key values for this data should be hardcoded values (i.e., **NOT** make use of sequences) and must consist of values below 100.
3. The entries and teams must be spread across at **least three carnivals** *which have been completed*. Within a given carnival at least two different events must be used.

For this task **ONLY**, you can look up and include values for the loaded tables/data directly where required. However, if you wish, you can still use SQL to get any non-key values.

You are reminded again that in carrying out this task you must not modify any data or add any further data to the tables which were previously populated by the supplied schema file.

[10 marks]

For <u>all subsequent tasks</u> (Task 3 onwards) **you are NOT permitted to manually**:

- manually lookup an attribute/s in the database to obtain *any* value,
- manually calculate values (including dates/times) external to the database, e.g. on a calculator and then use such values in your answers. ***ALL necessary calculations must be carried out as part of your SQL code***, or
- assume any contents in the database - rows in a table are potentially in a constant state of change

You must ONLY use the data as provided in the text of the questions. Where a particular case (upper case, lower case, etc.) for a word is provided you **must only use that case**. You may divide names such as Brigid Radcliffe into the first name of Brigid and a last name of Radcliffe if required. **Failure to adhere to this requirement will result in a mark of 0 for the**

relevant question.

TASK 3: Data Manipulation (10 marks)

For the following tasks, **your SQL must correctly manage transactions and use sequences to generate new primary keys for numeric primary key values** (under no circumstances may a new primary key value be hard coded as a number or value). Your answers for these tasks must be placed in the supplied SQL Script **T3-cr-dml.sql.**

(a) Create sequences (one for use per table) to provide primary keys for data entry into the following tables:

- COMPETITOR
- ENTRY
- TEAM

The sequences must begin at 100 and go up in steps of 1 (i.e., the first value is 100, the next 101, etc.).

[1 mark]

(b) Brigid Radcliffe, who has previously registered at City Run, has contacted the organisation and indicated she would like to run as a competitor in the "*CR Summer Series Melbourne 2024*" carnival. She would like to enter the "*21.1 Km Half Marathon*" event. You may assume that Brigid's phone number 1234567890, is unique in the current competitor data. She has indicated, for this carnival, that she will raise funds to support the "*Amnesty Internationa*l" charity.

Make these changes to the data in the database. This entire registration should be treated as a single transaction.

[3 marks]

(c) A few weeks later Brigid Radcliffe emails the City Run organisers and reports that she has suffered an injury in training and would like to change her entry (called downgrading) for the "*CR Summer Series Melbourne 2024*" carnival from the "*21.1 Km Half Marathon*" to the "*10 Km Run*". You may assume that Brigid's phone number 1234567890, is unique in the current competitor data. She also indicates that she would like to form a team called "*Kenya Speedstars*" for this carnival. City Run staff check and confirm that the team name is not currently in use for this carnival, so inform her that such a team can be created. The new team will support the "*Beyond Blue*" charity.

Make these changes to the data in the database. These changes must be treated as a single transaction.

[3 marks]

(d) The following week Brigid Radcliffe emails the City Run organisers and reports that her injury has got a lot worse and that she will need to withdraw from the "*CR Summer Series Melbourne 2024*" carnival. Although she has asked a few friends to join her team for this carnival she is not sure if any have actually taken up the offer; it is possible some have. She did direct that her team, "*Kenya Speedstars*", should be disbanded. Brigid Radcliffe indicated that she is looking forward to competing in the next 2024 carnival. You may assume that Brigid's phone number 1234567890, is unique in the current competitor data.

Make these changes to the data in the database. These changes must be treated as a single transaction.

[3 marks]

TASK 4: SQL Queries (46 marks)

Your answers for these tasks must be placed in the supplied SQL Script **T4-cr-queries.sql**

You are reminded that you must ***design your test data so that you always get output for your SQL queries and thus test them appropriately*** - this may require you to add further data as you move through completing the required tasks. Such extra data MUST be added as part of Task 2 (ie. as part of your load of test data). ***Queries that are correct but do not produce any output ("no rows selected" message) using your test data will lose 50% of the marks allocated***. Where a question indicates "Your output must have the form shown below" - this means the same appearance and alignment of columns/data as the sample output shows. Clearly your actual data may be different.

In any query where the sort order (descending or ascending) is not specified, you should use the system default.

(a) List the first name and the last name of the runners who have registered using a Gmail email address of the form '*gmail.com*' in carnivals organised by City Run.

The listing must include

- the date of the carnival(s),
- the name of the carnival(s),
- the event description of the event(s) joined for the carnival(s), and
- the first name and last name of the runners as a single column called fullname.

The listing should be displayed in ascending order of the carnival date and runner's full name within a carnival. Your output must have the *form* shown below.

CARNIVAL_DATE	CARNNAME	EVENTTYPEDESC	FULLNAME
Wed 01 February 2023	CR Summer Series Sydney 2023	21.1 Km Half Marathon	Sebastian Coe
Thu 06 April 2023	CR Autumn Series Brisbane 2023	5 Km Run	Fan Shu
Thu 06 April 2023	CR Autumn Series Brisbane 2023	5 Km Run	Ling Shu
Thu 06 April 2023	CR Autumn Series Brisbane 2023	5 Km Run	Nan Shu

[5 marks]

(b) List all registered runners who registered to support a charity as an INDIVIDUAL for the '*42.2 km Marathon*' event.

The listing must include
- the carnival date,
- the first name and last name of the runner as a single column called runner,
- the charity name,
- the charity contact person, and
- the full description of the supported event in which the runner is running

order the listing in ascending order of carnival date, within a carnival order by the charity name and then by the runners' full name within a supported charity.

[5 marks]

(c) List the number of events all competitors have completed over the previous two calendar years. For example if this report is run in 2023 it should show the events completed for 2022 (the last calendar year) and 2021 (two calendar years back). To allow this to occur dates must not be hardcoded as actual values.

The listing must include
- the competitors number,
- the competitors first name,
- the competitors last name,
- the competitors gender,
- how many events they entered two calendar years back,
- how many events they entered for the previous calendar year, and
- how many events they entered across these two previous calendar years. If they have entered no events for the two years, display 'Completed No Runs'

order the listing to show those who have completed no runs first, and for each of the two groups (those who have completed some runs and those who have completed no runs) by competitor number. Your output must have the form shown below.

COMPNO	COMPFNAME	COMPLNAME	COMPGENDER	TWOYRSBACK	LASTCALYEAR	LAST2CALENDARYEARS
18	Annamaria	Rose	F	0	0	Completed No Runs
19	Juan	Rose	M	0	0	Completed No Runs
20	Lynn	Nguyen	F	0	0	Completed No Runs
3	Brigid	Radcliffe	F	1	1	2
5	Sam	Ryan	M	1	1	2
1	Cathy	Freeman	F	0	1	1

[8 marks]

(d) List the total number of entries/participants in the "*5 Km Run*" for each carnival held in the year 2023.

The listing must include

- the date of the carnival,
- the carnival name, and
- the total number of entries/participants in the carnival for this event. Name the column as total_entries5Km

order the listing in descending order of total number of entries, if several carnivals have the same number of entries they should be ordered with the group in ascending carnival date order. Your output must have the form shown below.

CARNIVAL_DATE	CARNNAME	TOTAL_ENTRIES5KM
06-Apr-2023	CR Autumn Series Brisbane 2023	7
01-Feb-2023	CR Summer Series Sydney 2023	4

[8 marks]

(e) For all carnivals which have been run by City Run (i.e. the carnival has been finished), list those events which have had no entries.

The listing must include

- the date of the carnival,
- the carnival name,
- the event description

order the output by event description within the carnival date. Your output must have the form shown below.

CARNIVAL_DATE	CARNNAME	EVENTTYPEDESC
01-Feb-2023	CR Summer Series Sydney 2023	3 Km Community Run/Walk
06-Apr-2023	CR Autumn Series Brisbane 2023	10 Km Run
06-Apr-2023	CR Autumn Series Brisbane 2023	21.1 Km Half Marathon
06-Apr-2023	CR Autumn Series Brisbane 2023	3 Km Community Run/Walk

[8 marks]

(f) List the team details for each carnival where the most popular team name/s (the team name/s used most often across all carnivals) have been used.

The listing must include

- the team name,
- the date of the carnival,
- the team leaders competitor number as four digits and the first name and last name of the team leader as a single column called TEAMLEADER, and
- the number of members in the team

order the output by the team name and within the team name by carnival date.

Your output must have the form:

TEAMNAME	CARNIVALDATE	TEAMLEADER	TEAMNOMEMBERS
Happy Feet	01-Feb-2023	0015 Sebastian Coe	2
Happy Feet	08-Sep-2023	0004 Bob Ryan	4

[12 marks]

TASK 5: Design Modifications (13 marks)

Your answers for these tasks must be placed in the supplied SQL Script **T5-cr-mods.sql**

These tasks should be attempted *only* after task 2, 3 and 4 have been successfully completed. They are to be completed on the "live" database ie. the database with the data loaded from your previous work.

In completing this task, you **must**:

- if you need to add new columns, tables or related constraints, follow the naming conventions used in the data models and schema file which have been provided,
- provide column comments for any new columns that you add, and
- correctly manage any transactions used as part of your solution

(a) City Run organisers have noted that several competitors have enrolled in multiple events in the same carnival which they do not wish to occur.

Change the database structure to prevent a competitor from enrolling in two events for the same carnival.

[2 marks]

(b) City Run has decided that they would like to have the elapsed time (finish time - start time) for a runner in a particular event stored as part of the system, rather than having to calculate it every time it is required.

Add a new attribute which will record the runner's elapsed time in an event. The time should be stored as the number of minutes elapsed to two decimal places e.g. 26.12

This attribute must be initialised to the correct elapsed time based on the data which is currently stored in the system. You should note that the system may contain registrations for future events which currently do not have either a start or finish time.

[4 marks]

(c) When an emergency contact has been required to be contacted due to a problem with a competitor, there have been several instances of where the listed emergency contact has not been able to be contacted. To help alleviate this issue, City Run would like to be able to have a competitor, if they choose to do so, nominate more than one emergency contact.

Change the database structure to allow this to occur.

[7 marks]

TASK 6: Non Relational Database Queries - MongoDB (10 marks)

Your answers for this task (Task 6) must be placed in the supplied sql file **T6a-cr-json.sql** and the supplied MongoDB text file **T6b-cr-mongo.mongodb.js**

(a) Write an SQL statement in **T6a-cr-json.sql** to generate a collection of JSON documents using the following structure/format from the City Run tables used in tasks 4 and 5 (code/run this after task 5). Each document in the collection represents an event in a particular carnival and contains the entry details for that event. The document identifier (_id) is made up of the date of the event in the form ddmmyyyy and the event type code separated by an underscore e.g. "19092021_10K" :

```
{
    "_id": "19092021_10K",
    "carnival_date": "19-Sep-2021",
    "carnival_location": "The Tan, Birdwood Ave, Melbourne, 3004",
    "event": {
        "type": "10K",
        "description": "10 Km Run"
    },
    "no_competitors": 2,
    "competitors": [
        {
            "comp_no": 3,
            "name": "Brigid Radcliffe",
            "gender": "Female",
            "phone": "1234567890"
        },
        {
            "comp_no": 5,
            "name": "Sam Ryan",
            "gender": "Male",
            "phone": "0411222345"
        }
    ]
},
{
    "_id": "04092022_10K",
    "carnival_date": "04-Sep-2022",
    "carnival_location": "The Tan, Birdwood Ave, Melbourne, 3004",
    "event": {
        "type": "10K",
        "description": "10 Km Run"
    },
    "no_competitors": 3,
    "competitors": [
        {
            "comp_no": 1,
            "name": "Cathy Freeman",
            "gender": "Female",
            "phone": "0422666777"
        },
        {
            "comp_no": 5,
            "name": "Sam Ryan",
            "gender": "Male",
            "phone": "0411222345"
        },
        {
            "comp_no": 3,
            "name": "Brigid Radcliffe",
            "gender": "Female",
            "phone": "1234567890"
        }
    ]
},
```

Note only partial (two) documents shown.

[4 marks]

(b) Create a new collection named **events** and insert all documents generated in Task 6 (a) above into MongoDB in this collection.

[1 mark]

(c) List all events which have been inserted in (b)

[1 mark]

(d) List all events which have more than two participants and which are of type 10K. Display the carnival date, the location, the number of competitors and for all competitors registered their competitor number and name.

[2 marks]

(e) The carnival on the "*01-Feb-2023*" was moved to
"*Lake Gillawarna, Georges Hill, 2198*"
The location was not entered correctly in the database. Correct the necessary documents using a single MongoDB command. After correction use an appropriate db.find command to confirm the change was successful.

[2 marks]

SUBMISSION REQUIREMENTS

*Please note, if you need to resubmit, you **cannot** depend on your staffs' availability, for this reason, please be VERY CAREFUL with your submission.* **It is strongly recommended that you submit several hours before this time to avoid such issues.**

For this assignment there are seven files you are **required** to submit:

- T1-cr-ra.pdf
- T2-cr-insert.sql
- T3-cr-dml.sql
- T4-cr-queries.sql
- T5-cr-mods.sql
- T6a-cr-json.sql
- T6b-cr-mongo.mongodb.js

If you need to make any comments to your marker/tutor please place them at the head of each of your solution scripts/answers in the "Comments for your marker:" section.

Do not zip these files into one zip archive, submit seven independent SQL scripts. The individual files must also have been pushed to the FIT GitLab server with an appropriate history as you developed your solutions.

Late submission will incur penalties at the rate of -10 marks for every 24 hours the submission is late.

The seven files must also exist in your FITGitLab server repo and *show a clear history of development* (a **minimum** of two pushes per file).

Please note we **cannot mark any work on the GitLab Server**, you need to ensure that you submit correctly via Moodle since it is only in this process that you complete the required student declaration without which work **cannot be assessed**.

It is your responsibility to ENSURE that the files you submit are the correct files - we strongly recommend after uploading a submission, and prior to actually submitting, that you download the submission and double-check its contents.

Your assignment **MUST** show a status of "Submitted for grading" before it will be marked.

Submission status

Attempt number	This is attempt 1.
Submission status	Submitted for grading
Grading status	Not graded

If your submission shows a status of "Draft (not submitted)" it will not be assessed and **will incur late penalties after the due date/time**.

Please ***carefully*** read the documentation under the "Assignment Submission" on the Moodle Assessments page which covers things such as extensions and resubmission.

Marking Guide

Submitted code will be assessed against an optimal solution for these tasks. In some tasks where SQL is involved there are often several alternative approaches possible, such alternatives will be graded based on the code successfully meeting the briefs requirements. If it does, the answer will be accepted and graded appropriately.

Marking Criteria	Items Assessed
TASK 1 Relational Algebra Queries 6 marks	
	Maximum 6 marks - Satisfy brief requirements: • Marks awarded, as listed, (a) - (c) for relational algebra which meets the expressed requirement • Marks awarded for correct use of RA operations and the use of appropriate symbols • Mark penalty applied if your answer does not show an understanding of query efficiency ie. you must not make use of unnecessary joins, nor carry attributes and tuples up through the query which are not necessary • 0 marks will be awarded if SQL select statements are provided as solutions for task 1
TASK 2 Data Insert 10 marks	

Insert of required items of test data	**Maximum 5 marks - Insert of data:**
	
	• Marks awarded for correct insert of required data
	  ○ 20 ENTRY entries
	  ○ 5 TEAM entries
	• Marks awarded for correct management of transactions

Insert of valid test data	**Maximum 5 marks - Valid data inserted:** • Marks awarded for validity of data inserted

	○ meets the requirements expressed in the assignment brief ● Marks awarded for correct management of dates when inserting
Task 3 Data Manipulation 10 marks	
	Maximum 10 marks - Satisfy brief requirements: ● Marks awarded (a) - (d) for SQL code which meets the expressed requirement ● Mark penalty applied if commit not used appropriately ● Mark penalty applied if date handling and string database lookups not managed correctly
Task 4 SQL Queries 46 marks	
	Maximum 20 marks - Satisfy brief requirements: ● Marks awarded, as listed, (a) - (f) for SQL code which meets the expressed requirement ● Mark penalty applied if output does not match the form supplied for each question ● Mark penalty applied if date handling and string database lookups are not managed correctly ● Mark penalty applied if column aliases are not used when arithmetic calculation, concatenation, functions, or other output manipulation is used in a query ● Mark penalty applied if manual lookup/calculation is used for values from the database ● Statements which do not execute correctly in Oracle will be awarded a maximum of 50% of the available marks less 1 mark. For example, if a question is worth 6 marks and runs with an error in SQL the *maximum* mark awarded will be 2 marks
Task 5 Design Modifications 13 marks	
	Maximum 18 marks - Satisfy brief requirements: ● Marks awarded (a) - (c) for SQL code which meets the expressed requirement (including appropriate use of constraints). In making these modifications there must be no loss of existing data or data integrity within the database. ● Mark penalty applied if commit not used appropriately ● Mark penalty applied if column comments not used where required
Task 6 Non Relational Database Queries - MongoDB 10 marks	
	Maximum 16 marks - Satisfy brief requirements: ● Maximum of 4 marks awarded for creation of a JSON document which matches the supplied document format ● Marks awarded, as listed, (b) - (e) for MongoDB code which meets the expressed requirement

	• Mark penalty applied if field names and predicates (such as "&eq") are not enclosed in double quotes
Correct use of Git 5 marks	
	• Marks awarded for a minimum of fourteen pushes (two per file) showing a clear development history of the work for Assignment 2 • Marks awarded for correct Git author details used in pushes • Marks awarded for the use of meaningful commit messages (i.e. not blank or of the form "Push1")

Late submission penalty -5 marks for each 24 hours late or part thereof

Final Assignment Mark Calculation

Total: 100 marks, recorded as a grade out of 50